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IF estimation for time-frequency connected FM-CW signals based on signal separation using a combination of the Hough transform and time-frequency characteristic images

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Abstract

In practice, time-frequency connected frequency modulated-continuous wave (FM-CW) signals have been widely used, and it is significant to intercept such signals by estimating their instantaneous frequencies (IFs) under non-cooperative conditions. However, due to the small TF distance between the IFs of the FM and CW signal components, it may face the problem of treating the IF of the CW signal component as part of the FM signal component. To address this issue, this paper proposes a novel IF estimation algorithm based on signal separation using a combination of the Hough transform and TF characteristic images. First, the different TF characteristics of CW and FM signals are utilized to generate TF characteristic images and separate the CW and FM signal components. Then, the CW and FM signal components are identified and detected using a combination of the Hough transform and generated TF characteristic images. Finally, the IFs of the CW and FM signal components are estimated based on their detection results. Both experimental results based on simulations and sea trial data have verified the superiority of the proposed algorithm in terms of the accuracy in estimating the number of signal components and the normalized mean square error of IF estimation.

Introduction

To obtain both high range and Doppler resolutions, combined underwater acoustic pulse signals comprising both frequency modulated (FM) and continuous wave (CW) signal components have been widely used in active sonars [1]. In the combined signals, a special case exists: the time-

frequency (TF) distance between the ends of the instantaneous frequency (IF) curves of the FM and CW signal components is very small. This type of signal can be called a TF connected FM-CW (TFC FM-CW) signal, which is common in combined signals. For example, the signals emitted by the US Navy's surveillance towed array sensor system low-frequency and mid-frequency active sonars are such signals [[2], [3], [4]]. From the perspective of interception systems, estimating the IFs of such signals under non-cooperative conditions can facilitate the identification of signal components and the estimation of their frequency parameters, thereby supporting the underwater acoustic countermeasures. Therefore, research on the IF estimation of such signals is of practical significance. The TF distribution (TFD) peak detection is a classical algorithm for the IF estimation of underwater acoustic pulse signals [[5], [6], [7]]. However, if this classical algorithm is utilized to estimate the IFs of such a signal, the following problem arises: when the ending frequency of the FM signal is estimated, due to the small TF distance between the ends of the IF curves of the FM and CW signal components, the estimated IF of the CW signal component may be treated as if it were part of the IF of the FM signal component. This results in the CW signal component not being detected and the IF estimation of the FM signal component being inaccurate. The TFC FM-CW signal has similarities with the multi-component signal with intersecting IFs: the IF curves of at least two signal components can be approximated as intersecting, except that the intersection points coincide approximately with the ends of the IF curves. Therefore, it is easy to consider processing such signals using algorithms for the IF estimation of multi-component signals with intersecting IFs.

Many algorithms for the IF estimation of multi-component signals with intersecting IFs have been developed, such as ridge path regrouping (RPRG) [8,9], automatic directional ridge prediction tracker (ADRPT) [[10], [11], [12]], Radon-spectrogram [13,14] and chirplet transform [15,16]. By regrouping the TF ridge curves according to their variation rates at the intersections, RPRG can estimate IFs with unknown numbers, types, and time durations of signal components [8]. However, it cannot separate the FM and CW signal components for TFC FM-CW signals. By predicting the directions of the TF peaks, ADRPT obtains advantages similar to those of RPRG. However, it is not easy to estimate the directions of TF peaks quickly and accurately under low signal-to-noise ratio (SNR) conditions [17,18]. Moreover, for the TFC FM-CW signal, in the IF intersecting region, due to the interference of the CW and FM signal components in the IF intersecting region, the difference in the estimated directions of the TF peaks of CW and FM signal components is relatively small, making it difficult to separate the CW and FM signal components [10,19]. Due to the separability of the modes in the Radon domain, the Radon-spectrogram can also estimate the IFs using the inverse Radon transform. However, its estimation performance is relatively poor compared with other algorithms under low SNR conditions due to the difficulty in simultaneously separating different signal components and perfectly recovering signal components from the Radon domain [13,14]. Due to the separability of the modes in the time-frequency-chirp rate domain and the improvement in the resolution of the TFD, IF estimation algorithms based on chirplet transform, like frequency-chirprate reassignment (FCR) [20], group filter-matched CT-based signal separation (GFCT3S) [21], adaptive linear chirplet transform (ALCT) [15] and local maximum synchrosqueezing chirplet transform (LMSCT) [16], can accurately estimate IFs by extracting TF ridges in the time-frequency-chirp-rate domain under low SNR conditions with unknown numbers, types and time durations of signal components. However, the computational cost is much higher than that of most IF estimation algorithms because of the estimation in a three-dimensional space [15,22].

To solve the existing problems in estimating the IFs of TFC FM-CW signals using algorithms for the IF estimation of multi-component signals with intersecting IFs, a novel algorithm is proposed. The proposed algorithm involves three main steps. To begin with, the distinct TF characteristics of CW signals, FM signals and noise are utilized to generate TF characteristic images of CW and FM signals. These two images can separate the CW and FM signal components while mitigating noise effects. Following this, the CW and FM signal components are identified and detected by combining the Hough transform with their respective TF characteristic images. In the last, TF bands are set using the detection results, and IFs of the CW and FM signal components are estimated in the TF bands, which can avoid TF ridge tracking errors. The proposed algorithm not only separates the CW and FM signal components successfully and estimates IFs accurately but also reduces noise interference.

The remainder of this paper is organized as follows. The model of the TFC FM-CW signal and problem of estimating its IFs are presented in Section 2. The proposed algorithm is described in Section 3. In Section 4, simulations and experiments based on sea trial data are conducted to evaluate the performance of the proposed algorithm. Finally, the conclusions are given in Section 5.

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Section snippets

Signal model and problem formulation

The underwater acoustic pulse signal contaminated by marine environmental noise can be

modelled as follows [5]:
$$z(t) = \sum_{i=1}^I x_i(t) + v(t)$$

$$= \sum_{i=1}^I A_i \exp \left[j2\pi \int_{t_i}^t IF_i(r) dr + \varphi_{0i} \right] \text{rect}((t - t_i)/\tau_i) + v(t)$$
 where I

is the number of signal components, $x_i(t)$ is the i th signal component, $v(t)$ is the marine environmental noise which can be modelled as a combination of white Gaussian noise, colored noise and impulsive noise [23], A_i is the amplitude of the i th signal component, t_i and τ_i are the starting time and duration ...

Proposed algorithm

If the CW and FM signal components and noise can be separated, the IFs of both components can be estimated, thereby determining the TFC FM-CW signal IFs. The CW signal, FM signals, and noise exhibit distinct TF characteristics in their TF peaks, enabling separation. Therefore, to separate the CW signal components, the TF characteristics of CW signals can be evaluated to assign larger values to the TF peaks of the CW signal components and smaller values to the real TF peaks of FM signal ...

Simulations and experiments

In this section, the performance of the proposed algorithm is evaluated using both simulated and sea trial data. The IF estimation accuracy is quantified using two metrics: the accuracy in estimating the number of signal components (CR) and the normalized mean square error (NMSE) of IF estimation ($NMSE_{IF}$) [15]. They are calculated as follows: $CR = M_C / M \times 100\%$

$$NMSE_{IF} = 101g \left(\frac{1}{M} \frac{1}{N_t} \sum_{m=1}^M \sum_{i=1}^I \sum_{n=0}^{N_t-1} \left(\min \left\{ 1, \left(\frac{|IF_i^m(n\Delta t) - IF_i(n\Delta t)|}{\max\{IF_i(n\Delta t), \epsilon\}} \right)^2 \right\} \right) \right)$$
 where M is the number of Monte Carlo runs, M_C is the number ...

Conclusions

In this paper, a novel algorithm for estimating the IFs of TFC FM-CW signals based on signal separation using a combination of the Hough transform and TF characteristic images is proposed. Initially, the distinct TF characteristics of CW signals, FM signals, and noise are used to generate TF characteristic images of CW and FM signals, facilitating the separation of the CW and FM signal components and reducing noise interference. Subsequently, the CW signal components are detected, and FM signal ...

Credit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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